

Gradient as Causality

All Physical Processes as Energy-Distribution Homogenization

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Abstract:

Based on the core ideas of the author's series of manuscripts archived on Zenodo since 2025 and the complete 2026 paper Energy-Gradient-Driven Computational Universe, this article proposes a fundamental physical proposition: all forms of gradients are the ultimate drivers of all physical processes, and the inevitable consequence of gradient-driven evolution is the homogenization of distributions. From temperature gradients in heat conduction, pressure gradients in fluid dynamics, electrochemical potential gradients in chemical reactions, to action gradients in gravitational and quantum processes, all are essentially reducible to differences in energy distribution across the Space Elementary Quantum (SEQ) network—namely, energy gradients. This article argues that the energy gradient is the sole primitive "causal driver," determining the state transition paths of the spacetime network and leading to entropy increase and thermodynamic equilibrium. Importantly, the model emphasizes that actual quantum processes are inherently dissipative. The so-called "reversible" or "unitary" quantum evolutions observed in experiments are maintained only by continuous energy input from external components such as lasers, magnetic traps, or other control devices. When these external apparatus are included in the total isolated system, the overall evolution remains irreversibly dissipative. This perspective is highly consistent with the framework proposed by Arto Annala and others, in which energy gradients serve as the fundamental driver of evolution, and provides a unified and intuitive physical picture for understanding the arrow of time, life phenomena, and large-scale cosmic evolution.

Keywords: Energy gradient; Causality; Entropy increase; Least action; Space Elementary Quantum; Evolutionary driver; Homogenization of distribution; Thermal dissipation

1. Introduction: From Phenomena to Essence – Gradients Are Everywhere

In natural phenomena and classical physical observations, "gradients" are ubiquitous. Heat flows from high to low temperature, driven by a temperature gradient. Water flows from high to low elevation, driven by a gravitational potential gradient (or pressure gradient). Electric current flows from high to low electric potential, driven by an electric potential gradient. Matter diffuses from high to low concentration regions, driven by a chemical potential gradient. These seemingly diverse physical processes all follow the same underlying logic: a non-uniform distribution generates a gradient, the gradient drives a flow or change, and the flow tends to eliminate the gradient, thereby making the distribution more uniform.

The result of gradient-driven evolution is the progression of a system from a non-equilibrium state toward equilibrium. This is the most intuitive manifestation of the second law of thermodynamics in nature. From the diffusion of a drop of ink into water to the release of nuclear energy inside stars and its radiation into surrounding space, all are homogenization processes driven by energy gradients.

It is particularly important to note that the SEQ model underlying this article explicitly asserts that all real quantum processes are inherently thermodynamically dissipative. Although in laboratories we can use laser cooling, magneto-optical traps, microwave pulses, and other external means to make quantum systems exhibit approximately reversible unitary evolution, this "reversibility" is not an intrinsic property of the system itself, but rather the result of continuous work input and energy injection from external devices. Once the lasers, power supplies, control systems, and other external components are included in the total isolated system, the overall energy distribution still follows the rule of one-way flow from higher-energy regions to lower-energy regions, and the total entropy of the system continues to increase irreversibly. In other words, there is no truly isolated, spontaneously reversible quantum process. This recognition poses a profound reexamination of the status of "unitarity" as a fundamental postulate of quantum mechanics.

2. All Gradients Are Reducible to Energy Gradients

Although different branches of physics use different types of "potentials" to describe gradients (temperature, pressure, electric potential, chemical potential, etc.), from a microscopic viewpoint, all these gradients originate from non-uniform distributions of energy in space. In the "Energy-Gradient-Driven Computational Universe" model, space consists of discrete "Space Elementary Quanta" (SEQ), each carrying an integer multiple of Planck's constant as its energy. The state of the system is uniquely determined by the energy distribution across all SEQs.

From this perspective:

A temperature gradient is essentially a spatial difference in the average kinetic energy (i.e., energy) of molecules.

A pressure gradient is a spatial variation in molecular number density and kinetic energy density.

An electric potential gradient is a spatial variation in the interaction energy between charges.

A chemical potential gradient is a spatial difference in exchange energy between particles.

Therefore, all classical and modern physical driving phenomena can be reduced to spatial gradients of energy distribution. The fundamental dynamical rule of the model is precisely: energy always flows from higher-energy SEQ to adjacent lower-energy SEQ, one quantum of Planck's constant at a time. This simple rule drives all changes in the universe.

3. Gradient-Driven Dynamics and Causality

At the philosophical level, causality is usually expressed as "cause produces effect." At the physical level, this article argues that the real "cause" is the existence of an energy gradient, and the "effect" is the process by which the energy gradient is partially or completely eliminated. That is, the gradient is the cause, the flow is the process, and homogenization is the result.

This understanding unifies multiple branches of physics:

In Newtonian mechanics, force is the negative gradient of potential energy ($F = -\nabla U$), so acceleration is a consequence of gradient driving.

In thermodynamics, entropy increase is precisely the process in which energy gradients continuously decrease until they vanish.

In quantum mechanics, the phase gradient of the wavefunction drives probability flow, and the principle of least action in path integrals can be understood as the system preferentially selecting the path with the steepest descent of the action gradient among all possible paths.

In general relativity, the distribution of matter creates gradients in the spacetime metric, thereby affecting the frequency of light and the geodesic motion of particles.

As pointed out by Tuisku, Pernu, and Annala (2009), both quantum behavior and macroscopic evolution can be understood in a unified framework where energy

gradients serve as the fundamental driving force. Annala (2025) further demonstrated that from the quantum of action to the second law of thermodynamics, gradient-driven dynamics are universal and irreversible.

4. From Gradients to Entropy – The Inevitability of Homogenization

In the referenced model, system evolution follows a core principle: energy is preferentially transferred along the maximum adjacent energy gradient. Through this rule, the system's energy distribution gradually becomes more uniform. The author introduces a "multiplicative entropy" to quantify this process, defined as the product of all SEQ energy values. This entropy monotonically increases with each energy transfer until it reaches its maximum when all SEQ energies are equal. Thus, entropy increase is not a statistical approximation but an exact mathematical description of the homogenization process in a discrete spatial network.

This implies that the inevitable end point of gradient driving is complete homogenization—the thermodynamic equilibrium state. At this point, no global gradient exists, and all macroscopic evolution ceases. This is the microscopic dynamical essence of "heat death."

5. Conclusion: Gradients as the Underlying Causal Driver of Cosmic Evolution

In summary, this article derives the following core propositions:

All physical processes are driven by some form of gradient.

All gradients are essentially manifestations of energy gradients at corresponding levels.

The only direction of gradient driving is to eliminate itself, i.e., to achieve homogenization.

Gradients are the starting point of real physical causal chains; they are the physical carriers of "cause."

Real quantum processes are inherently irreversible and dissipative; experimental reversibility is maintained by external work input.

Therefore, we can state: Gradients are causality, and the homogenization of energy distribution driven by energy gradients is the underlying driving mechanism of cosmic evolution. The universe is not propelled by "forces" or "laws" but by the non-uniformity of energy distribution—namely, gradients—and evolves irreversibly toward uniformity. This viewpoint provides a solid philosophical and model-based foundation for a unified picture of physics and cosmology.

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